

Decision

Decision on changes to the financial parameters of the cap and floor regime for Window 3 electricity interconnectors

Publication date:	3 December 2024
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In September 2023 we published a consultation (the **First Consultation**) on changes to the financial parameters of the cap and floor regime for Window 3 electricity interconnectors and risk considerations for offshore hybrid assets. Subsequently, in July 2024, we published a further consultation on the cap rate for the cap and floor regime for Window 3 electricity interconnectors (the **Second Consultation**).

This document addresses Section 1 of the First Consultation (Proposed changes to the cap and floor regime design parameters for Window 3 interconnector projects) and the entirety of the Second Consultation. It provides our decision on the financial parameters that will apply in the cap and floor regime for Window 3, and also for possible application to subsequent application windows¹, electricity interconnectors.

The feedback in the responses to Section 2 of the First Consultation (Risk considerations for offshore hybrid assets) has been taken into account in the consultation¹ on further detailed regime parameters for the Offshore Hybrid Asset pilot scheme, published in September 2024.

¹ Ofgem (2024), [Consultation on further detailed regime parameters for the Offshore Hybrid Asset pilot scheme](#)

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Executive Summary

Introduction

We have outlined our decision on financial parameters for Window 3, and for possible application to interconnector projects in any future application windows, with supporting detail in Sections 2 and 3 of this document. This decision addresses the topics covered in Section 1 of the First Consultation and the entirety of the Second Consultation.

Summary of decisions on changes to the financial parameters for third application window (Window 3) interconnector projects

The decisions, following our analysis of the consultation responses to our proposals, are summarised below.

- **Inflation metric:** We have decided to remove the Retail Prices Index (**RPI**) as an inflation index from the cap and floor regime design and adopt the Consumer Prices Index including owner occupiers' housing costs (**CPIH**) as the primary inflation measure. We have decided to use the Consumer Prices Index (**CPI**) as a proxy for CPIH until CPIH forecasts are reliably available.
- **Cost of debt:** We have decided to base our approach for calculating rates of return at the floor on a 20-day average of 50% weighting to the iBoxx GBP Non-Financials 15yr+ A rated index and a 50% weighting to the iBoxx GBP Non-Financials 15yr+ BBB rated index and to determine the rates at Final Investment Decision (FID) or Financial Close (FC), as appropriate. To convert nominal to real yields we have decided to use CPI inflation target.
- **Asset beta and notional equity beta:** We have decided to use an asset beta of 0.60 for the third application window for interconnectors. This figure is derived from an average of the six comparators' asset betas - Drax, SSE, Iberdrola, Orsted, RWE and National Grid using equal weighting and a 5-year estimation window. We have decided to set the notional equity beta at 1.125, calculated using a 50% notional gearing and a debt beta of 0.075.
- **Risk-free rate (RFR):** We have decided to calculate the RFR using the 20-day average of 20-year Index Linked Gilts (ILG) yield at FID or FC, as appropriate. To convert RPI-real to CPIH-real yields we have decided to apply RPI-CPI wedge.
- **Total market return (TMR):** We have decided to align the TMR with the range estimated using the Ofgem's latest thinking regarding RIIO price controls for gas and electricity transmission networks, as updated periodically and to use the midpoint of this range.

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- **Equity transaction cost:** We have decided to retain the equity transaction cost allowance as 5% in the default regime.

Details of the proposals, stakeholder responses and the rationale for our decisions are provided in Sections 2 and 3. In Section 4, we provided a summary of the changes to the financial parameters that we will use to calculate returns under the cap and the floor regime for Window 3 interconnectors, and to potentially apply to successful projects in any subsequent application windows. This structure ensures clarity and transparency in our decision-making process, allowing stakeholders understand the basis of our adjustments and the expected impact on financial outcomes.

Next steps

The decisions in this document will inform the development of a revised version of the Cap and Floor Regime Handbook for Window 3 projects. These decisions will also be used in the subsequent stages of assessment and approval for those Window 3 projects that have already received our in-principle approval for a cap and floor regime in our Initial Project Assessment (**IPA**) decision.² The decisions in this document potentially also to apply to future application windows for cap and floor interconnector projects.

² Ofgem (2024), [Decision on the Initial Project Assessment of the Third Cap and Floor Window for Electricity Interconnectors](#)

1. Introduction

Background

- 1.1 Our Interconnector Policy Review decision of December 2021 noted our intention to run Window 3 for electricity interconnectors.³ This window was open from 1 September 2022 to 10 January 2023. We received seven applications for Window 3, proposing connections to Belgium, Denmark, France, Germany, the Netherlands, Northern Ireland and the Republic of Ireland.

The cap and floor regime for electricity interconnectors

- 1.2 The cap and floor regime is the regulated route for electricity interconnector development in Great Britain (**GB**). It is a market-based approach which aims to incentivise developers to deliver interconnector capacity by limiting developers' exposure to electricity market price risk. The regime sets a yearly maximum (cap) and minimum (floor) level for the revenues that the interconnector can earn over a 25-year period.⁴ Revenues above the cap are passed back to network users, benefitting consumers, while revenues below the floor are topped-up by consumers.
- 1.3 Project-specific cap and floor levels are set based on efficient project costs using a Regulatory Asset Value (**RAV**) model. The cap and floor levels are built from building blocks of capital costs, operating and maintenance costs, decommissioning costs, tax and allowed return. The cap and floor levels are then profiled so that they are flat in real terms over the duration of the regime.
- 1.4 The allowed return at the cap is calculated by applying a rate of return set to reflect a maximum permitted cost of equity, to 100% of the RAV. The allowed return at the floor is calculated by applying a rate of return set to reflect the cost of debt to 100% of the RAV.
- 1.5 This decision follows the consultation on changes to the financial parameters of the cap and floor regime for Window 3 electricity interconnectors and risk

³ Ofgem (2021), [Interconnector Policy Review - Decision](#)

⁴ Ofgem (2021), [Interconnector Cap and Floor Regime Handbook](#), page 6.

considerations for Offshore Hybrid Assets⁵ (the **First Consultation**) which we published in September 2023 and the further consultation on the cap rate for the cap and floor regime for Window 3 electricity interconnectors⁶ (the **Second Consultation**) which we published in July 2024. These consultations address the approach to the financial parameters used to set the cap and floor levels for licensees.

Background for estimating allowed rates of return for the third window projects

- 1.6 Our First and Second Consultations flagged that we intend to align, where appropriate, our methodology for setting financial parameters for Window 3 of the cap and floor regime with other regulatory regimes applied by Ofgem and regimes applied by other utility regulators in GB.
- 1.7 We also flagged that we would incorporate the relevant UK Regulators Network (**UKRN**) Guidance⁷ recommendations with regards to estimating the cost of capital into our methodology.

Context and related publications

- 1.8 The First Consultation had two parts: firstly, on changes to certain financial parameters of the cap and floor regime for Window 3 interconnector projects, and; secondly, risk considerations for offshore hybrid assets (**OHA**s), in particular the Non-Standard Interconnector projects of the OHA pilot scheme. The feedback in the responses to the First Consultation on risk considerations for OHAs has been taken into account in the consultation⁸ on further detailed regime parameters for the OHA pilot scheme and is not considered in this document.
- 1.9 We asked the following questions in Section 1 of the First Consultation:

⁵ Ofgem (2023), [Consultation on changes to the financial parameters of the cap and floor regime for Window 3 electricity interconnectors and risk considerations for Offshore Hybrid Assets](#).

⁶ Ofgem (2024), [Further consultation on the cap rate for the cap and floor regime for Window 3 electricity interconnectors](#).

⁷ UKRN (2023), [UKRN guidance for regulators on the methodology for setting the cost of capital](#).

⁸ Ofgem (2024), [Consultation on further detailed regime parameters for the Offshore Hybrid Asset pilot scheme](#)

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Q1. Do you agree with our proposal to move away from the use of RPI to CPIH as the inflation measurement for the cap and floor regime, and is there a better proxy to use for CPIH than CPI?

Q2. Do you agree with the changes we are proposing to introduce to the way we calculate the floor rate? If not, could you please explain why and provide evidence for your reasons?

Q3. Do you have any alternatives to our proposed changes? If so, could you please elaborate on them and present evidence on the potential impact these might have on the current floor rate?

Q4. Do you agree with the issues raised and the proposed changes to the cost of equity? If not, could you please explain why and provide evidence for your reasons as well as provide alternatives?

Q5. Do you agree with our analysis on impacts related to the risk-reward balance?

Q6. Do you agree with CEPA that the equity transaction costs should not be higher than 3%? If not, could you explain why and provide evidence for your reasons?

1.10 We asked the following questions in the Second Consultation:

Q1. Do you agree with our methodology for calculating equity beta for Window 3 interconnector projects? If not, could you please explain why, provide evidence for your reasons and suggest alternatives?

Q2. Do you agree with the comparators we are proposing to use to calculate the beta parameter? If not, could you please explain why, provide evidence for your reasons and suggest alternative methods?

Q3. Do you agree with the proposed approach for determining the Total Market Return parameter? If not, could you explain why, provide evidence for your reasons and suggest alternative methods?

1.11 The following other Ofgem publications are relevant to this decision.

[Application Guidance for the Third Cap and Floor Window for Electricity Interconnectors | Ofgem](#)

[Consultation on changes to the financial parameters of the cap and floor regime for window 3 electricity interconnectors and risk considerations for offshore hybrid assets | Ofgem](#)

[Consultation on the Initial Project Assessment of the Third Cap and Floor Window for Electricity Interconnectors | Ofgem](#)

Our decision-making process

- 1.12 We will consolidate the changes to the cap and floor regime for Window 3 in an updated Cap and Floor Regime Handbook applicable to Window 3 projects. This will include changes arising from this decision, the decision on timelines and incentives⁹, and the decision on indexation for Interest during Construction¹⁰. The decisions in this document potentially also apply to any potential future application windows for cap and floor interconnector projects.
- 1.13 Our consultation on the IPA for the Window 3 projects was published in March 2024, and we published our IPA decision on 12 November 2024¹¹.

General feedback

- 1.14 We believe that consultation is at the heart of good policy development. We are keen to receive your comments about this report. We'd also like to get your answers to these questions:
1. Do you have any comments about the overall quality of this document?
 2. Do you have any comments about its tone and content?
 3. Was it easy to read and understand? Or could it have been better written?
 4. Are its conclusions balanced?
 5. Did it make reasoned recommendations?
 6. Any further comments

Please send any general feedback comments to stakeholders@ofgem.gov.uk.

⁹ Ofgem (2023), [Decision on Timelines and Incentives changes for the Third Cap and Floor Window for Interconnectors](#)

¹⁰ Ofgem (2024), [2024-25 IDC Decision Letter](#)

¹¹ Ofgem (2024), [Decision on the Initial Project Assessment of the Third Cap and Floor Window for Electricity Interconnectors](#)

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2. General considerations

Introduction

- 2.1 We received certain general feedback on the consultation on the extent of the consultation and whether project financed (i.e. limited recourse) interconnector projects should be permitted higher returns than those financed on balance sheet as part of a portfolio. We address these points in this section before proceeding to more detailed matters in Section 3.

Consultation feedback

- 2.2 One stakeholder commented that “a piecemeal and late review of the window 3 cap and floor could undermine investment in interconnectors and wider energy infrastructure”. They went on to say that:

“While we acknowledge the need to develop and adapt the interconnector cap and floor regime over time (for example, we recognise the need to review the beta value that was set in 2014), it is fundamentally flawed to adjust specific parameters within the cap and floor in isolation without reviewing the cost of equity and cost of debt more broadly. Ofgem’s Interconnector Policy Review reviewed the broader interconnector cap and floor regime a few years ago, in contrast this consultation just considers a few parameters without considering the fuller impact on cost of equity. We consider that any reviews of the cap and floor regime, or any regulatory financial model, need to be done ‘in the round’ and consider all parameters together. In the absence of a more robust review, and where there is still considerable uncertainty around different parameters, we believe the interconnector cap and floor regime should stay as is given the need for the regime to stay competitive internationally, attracting investment necessary to enable new electricity infrastructure projects.”

- 2.3 Another stakeholder recommended a broader set of changes to the cap and floor regime to recognise challenges faced by project-financed interconnector developers in finalising the parameters for Window 3, as follows:

- “1. Adjusting the equity beta to better reflect the specific risks associated with single-asset, project-financed interconnectors.
2. Ensuring that the cap rate is set at a level that balances consumer protection with the need to attract investment.

3. Introducing flexibility within the cap and floor regime to account for changes in the market and operating environment.
4. Providing specific provisions or adjustments for single-asset developers to ensure that the regime remains viable and supportive.
5. Considering the broader policy goals of the energy transition when setting the cap and floor levels to support the timely development of interconnector projects.”

Ofgem response to consultation feedback

- 2.4 The overall cap and floor regime was comprehensively reviewed in the ICPR decision of December 2021, and we do not believe that a further such review is necessary at this stage. Prior to publishing the First and Second Consultations, we conducted a thorough evaluation of all regime parameters and identified the key areas that required attention, addressing these appropriately in the consultations.
- 2.5 In the First and Second Consultations we have considered the key regime parameters applicable during the operational phase of the cap and floor regime, and also the equity transaction cost which is added to the RAV at the start of the operations period. Previously, we reviewed and updated the interest during construction methodology for cap and floor interconnectors.
- 2.6 The cap and floor regime is developer-led and as such, we continuously review its regime parameters during the application windows to address the nature and volume of applications. Key investment decisions, including later development and construction expenditures, are yet to take place because successful Window 3 projects have only recently had their IPA decisions. In addition, the consultation process on regime parameters has been ongoing since September 2023.
- 2.7 The responses to the First Consultation in respect of the beta parameter did not suggest a robust set of comparator companies and so we undertook the Second Consultation. We consider it to be more efficient to focus on parameters where there is need for review than consulting on all of the regime parameters.
- 2.8 Regarding the suggested broader set of changes to the cap and floor regime when finalising the parameters for Window 3 to respond to challenges faced by project-financed interconnector developers (see paragraph 2.3 above), we consider that our approach of selecting suitable comparator companies for the beta parameter, applying the cap cost of finance to 100% of the RAV, and providing full indexation of the cap, strikes a balance that protects consumers and

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takes account of investor's needs. We do not consider it appropriate at this time to go beyond the scope of the changes to the regime parameters proposed in the First and Second Consultations and described in this decision.

3. Changes to the cap and floor regime design financial parameters for Window 3 interconnector projects

Background

- 3.1 In the First Consultation published in September 2023, we proposed the methodology for setting allowed return parameters for Window 3 projects of the cap and floor regime. As a part of our consultation, we sought stakeholders' feedback on the appropriate inflation metric and the financial parameters to use when calculating the cost of debt and the cost of equity for Window 3.
- 3.2 This was followed by a Second Consultation published in July 2024, where we presented our analysis and proposed a specific methodology for estimating the equity beta parameter in the third window. We also updated the previously proposed methodology for estimating TMR and addressed issues regarding our proposed methodology for calculating RFR.
- 3.3 We consider aligning with the UKRN Guidance, where this is relevant, beneficial for both regulatory transparency and predictability, and where the aligning will ultimately serve the interests of consumers. We will incorporate best practice from the UKRN Guidance recommendations when estimating the financial parameters for the third window projects, except where there are good reasons to deviate from these guidelines.

Inflation metric

Consultation summary

- 3.4 For the projects within the first and the second application windows we use the RPI as the inflation index for the cap and floor regime to achieve the following:
- Set the cap, floor levels and interconnector Interest During Construction (**IDC**) rates in real (stripped of inflation) terms.
 - Adjust the cap and floor levels, as set at Post Construction Review stage, to reflect outturn inflation. To compare those levels against the actual revenues, we multiply them by an index factor to express them in nominal prices for each year of the regime.
- 3.5 In our First Consultation we proposed to move from the use of RPI to CPIH as the inflation measurement for the cap and floor regime and to use CPI as a proxy for CPIH until CPIH forecasts are reliably available. Our proposed approach aligns

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with the wider Ofgem regulatory regimes. In the RIIO-2¹² Final Determinations, for the transmission and gas distribution price controls that commenced in 2021 we decided to implement a change from RPI to CPIH from 1 April 2021 for RIIO-2.¹³

- 3.6 We proposed to use CPI forecasts from credible sources such as the Office of Budget Responsibility (**OBR**) and HM Treasury, as a suitable proxy for estimating CPIH until reliable CPIH-based estimates are available.
- 3.7 We asked stakeholders if they support our proposal to transition from using RPI to CPIH as the inflation measurement for the cap and floor regime, and if they believe there is a more suitable proxy for CPIH than CPI.

Summary of consultation responses

- 3.8 Most stakeholders consider the transition from RPI to CPIH to be reasonable.
- 3.9 A stakeholder noted that some costs that interconnection projects have to bear, for example land rent, are indexed based on RPI or other methodologies. The stakeholder suggested that there is a need for a carve-out allowing the implementation of RPI or another specific adjustment mechanism for the purposes of the floor to account for the special nature of certain costs.
- 3.10 A stakeholder raised a concern around the ability to hedge the inflation at Financial Close as a result of not having sufficiently deep and liquid markets, particularly given the very large size of the project financing.
- 3.11 A stakeholder highlighted the use of RPI ILG in the determination of the RFR with respect to the cost of equity. The stakeholder stated it would be sensible to maintain uniformity of one inflation index to all cap & floor calculations and continue to utilise RPI until the RPI methodology is aligned to CPIH from 2030. They stated that they consider CPIH to better reflect an inflation index for interconnectors than CPI.
- 3.12 A stakeholder stated that they were not aware of a better general proxy for CPIH than the OBR's forecast of CPI over the relevant timescales.

¹² RIIO stands for 'Revenue = Incentives + Innovation + Outputs'

¹³ Ofgem (2020), [RIIO-2 Final Determinations for Transmission and Gas Distribution network companies and the Electricity System Operator](#), [RIIO-2 Final Determinations – Finance Annex \(REVISED\)](#), page 111.

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- 3.13 A stakeholder noted that CPI appears to be used more widely in financial markets than CPIH, which would allow developers to be able to better manage the future inflation risks. The stakeholder also cited the absence of forecasts for CPIH or a wedge between CPI and CPIH and recommended that CPI indexation rather than CPIH is adopted.
- 3.14 A stakeholder requested clarity on how forecasts for CPIH will be derived. They highlighted that the current wedge between CPI and CPIH is relatively significant and queried how Ofgem intends to reflect this difference. The stakeholder requested further assurance that the transition will be net present value neutral. They suggested that CPI may be a better option than CPIH.
- 3.15 A stakeholder stated there was insufficient evidence presented in the consultation to show the impact of the proposed change. The stakeholder added they would expect Ofgem to show the potential impact on cap and floor levels.

Decision and rationale

Financial parameter	Decision
Inflation metric	We have decided to remove the RPI as an inflation index from the cap and floor regime design and adopt the CPIH as the primary inflation measure. We have decided to use the CPI as a proxy for CPIH until CPIH forecasts are reliably available.

- 3.16 We note the broad support for our proposal to move our inflation metric from RPI to CPIH.
- 3.17 In 2018 the Office for National Statistics (**ONS**) stated that RPI is no longer seen as a credible measure of inflation and that the methods used to produce it are not consistent with internationally-recognised best practice, a shortcoming that led to it losing National Statistics status in 2013.¹⁴ We see benefits in adopting CPIH, given a wider government objective that it will become the headline measure over time.
- 3.18 We aim to address the current lack of CPIH forecasts by utilising medium-term forecasts of CPI from reputable sources, such as the OBR, as a proxy until such

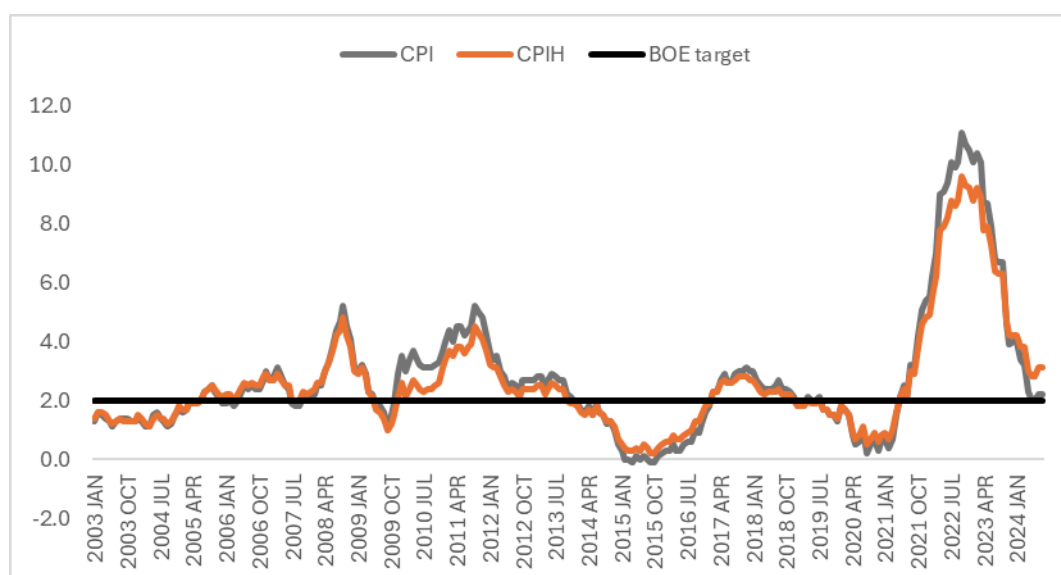
¹⁴ ONS (2018), [Shortcomings of the Retail Prices Index as a measure of inflation - Office for National Statistics](#).

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time as reliable CPIH forecasts are available. Historical CPI and CPIH rates of inflation have been very close on average. This can be seen in Figure 1. The average difference between CPIH and CPI over a longer-term dataset is only 0.04%, and it is far from certain what the magnitude or direction of any future difference between the measures would be.¹⁵ Therefore, we consider assuming that CPI is a close proxy for CPIH is appropriate.

- 3.19 We understand the recent short-term differential between CPI and CPIH was predominately driven by energy price volatility and other transient macroeconomic factors and the differing weightings of inputs between the two indices most affected by these factors. We are not aware of evidence to suggest a structural wedge between the two indexes and so do not consider that a forward adjustment is merited to CPI forecasts in order to derive a CPIH equivalent.

Figure 1: History of CPI and CPIH inflation rate 2003 - 2024



Source: Ofgem's analysis of ONS data

- 3.20 With respect to a long-term estimate of inflation, we had considered using the OBR 5th year CPI conditional forecast in line with RIIO-2. The 5th year OBR forecast has usually aligned to 2% but recent¹⁶ (conditional) forecasts from the OBR have been below 2%. We consider it unlikely that long run CPIH inflation

¹⁵ Ofgem (2024), [RIIO-3 Sector Specific Methodology Decision – Finance Annex](#), pages 60-61.

¹⁶ OBR (2023), Economic and fiscal outlook, March 2023, page 144.

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expectations would reside below the Bank of England CPI target, therefore, for the purpose of long-term inflation forecast we will adopt the Bank of England CPI inflation target (2%).¹⁷ This is in line with the approach proposed for RIIO-3.

- 3.21 We aim to maintain uniformity of one inflation index to all cap and floor calculations within the third application window including the cost of equity and IDC. Within the cost of equity methodology, we adjust RPI-real ILG yields to 'CPIH-real' terms by estimating the difference between future CPIH and RPI inflation - often referred to as the inflation 'wedge'. In March 2024 we published the Decision on the consultation to modify the inflation index used in the calculation of IDC¹⁸ in which we decided to move from the RPI to CPIH and to use the consumer prices index (CPI) as a proxy for CPIH, until CPIH-based forecasts become reliably available. This decision applies to Window 3 cap and floor electricity interconnectors.
- 3.22 A stakeholder noted that some costs that interconnection projects are exposed to are indexed based on RPI or other methodologies. Controllable opex and replacement expenditure allowances are set in real terms. At the present time, we do not consider there is evidence to suggest the price base change would, in-the-round, generate significant risk that the overall allowances would become misaligned to actual efficient costs due to differences in the general index and the underlying inflation rates associated with each cost item. We consider many of the examples cited by the stakeholder would be categorised as non-controllable opex and, therefore, be treated as a pass through. We do not consider the indexation approach to be economically relevant with respect to pass through costs.
- 3.23 Regarding stakeholder concerns relating to the ability to conduct hedging at financial close, we note that markets evolve over time and the potential impact of any limitation in availability of inflation hedging products on projects is uncertain. We will closely monitor the available evidence over time and will consider, in the

¹⁷ Since 2003, when the Bank of England adopted CPI inflation targeting, the Year-on-Year change of CPIH has averaged 2.6%.

¹⁸ Ofgem (2024), [Decision on the consultation to modify the inflation index used in the calculation of Interest During Construction \(IDC\) and the IDC rates to apply during 2024-25 for offshore transmission projects and Window 3 cap and floor electricity interconnectors](#).

context of the consumer interest, whether adjustments to the approach are required in future.

- 3.24 Some stakeholders questioned the value neutrality of this change. In our view, this decision, to move away from RPI to CPIH, is broadly neutral for both investors and consumers. In addition, given the change in 2030, the potential for any mismatch in outcome for Window 3 projects, given the connection dates of these projects prior to the end of 2032, under RPI and CPIH is very small.
- 3.25 We consider sufficient evidence has been provided in respect of the potential impact on cap and floor levels. We refer stakeholders to Section 4 for a comparison in cap and floor rates between Window 3 and Window 2. We will be developing a methodology to reflect the change in inflation index for use within the Cap and Floor Financial Models.

Approach for estimating floor rate for Window 3

Cost of debt

Consultation summary

- 3.26 Rates of return at the floor for projects within first and second application windows are calculated based on our estimate of a reasonable cost of debt which we calculate using the 20-day average of iBoxx GBP Non-Financials 10yr+ indices with credit ratings of A and BBB which are equally weighted. The 20-day average is calculated by using 20 working days ahead of the FID or FC date with FID or FC date being the last day of the data sample. Each daily datapoint is converted to real terms with 10-year breakeven RPI data published by the Bank of England.
- 3.27 For the third window we proposed to change the indices on which we base our calculations from iBoxx GBP Non-Financials 10yr+ to iBoxx GBP Non-Financials 15yr+. We supported our proposal stating that it would enable aligning the maturity length of the yield indices more closely with the operational period of the cap and floor regime, which is 25 years.
- 3.28 In the First Consultation we stated that modifying the inflation metric would impact the cost of debt calculations.
- 3.29 In the First Consultation we asked stakeholders whether they agree with the changes, we proposed to introduce to the way we calculate the floor rate. We also asked whether stakeholders have any alternatives to our proposed changes.

Summary of consultation responses

- 3.30 Most stakeholders were supportive of our proposed change to base our calculations on iBoxx GBP Non-Financials 15yr+ rather than on iBoxx GBP Non-Financials 10yr+ index and considered it reasonable.
- 3.31 A stakeholder requested that Ofgem attributes a higher cost of debt and equity for projects sponsored on a standalone basis compared to those funded on balance sheet by large corporates. Some stakeholders questioned whether the changes would impact the ability to apply for a project finance specific variation to the floor rate.
- 3.32 A stakeholder stated that Ofgem should consider alternative indices that would more accurately capture the yields of longer maturities.
- 3.33 Some stakeholders noted that there is lack of clarity on how Ofgem intends to deflate the 20-day trailing average from nominal to CPIH-real terms.
- 3.34 A stakeholder requested further clarity on how breakeven inflation curves will be applied with respect of the new methodology. The stakeholder suggested that the deflator could be calculated from RPI as a backup if a certain threshold was triggered.
- 3.35 Another stakeholder requested clarity as to the purpose of switching the floor calculation to a high and low range. The same stakeholder noted that, all else equal, having a range as opposed to a point estimate introduces more uncertainty into the regime.
- 3.36 A stakeholder suggested that adopting a CPI methodology would be more desirable given the availability of forecasts for this measure.
- 3.37 A stakeholder expressed concerns that the proposed changes will impact the calculation of the floor rate for those projects that rely on the actual cost of debt, as permitted under variations to the special conditions to facilitate project financing.
- 3.38 A stakeholder suggested that the proposed change would create an unfair advantage for W3 interconnectors over W2 interconnectors.
- 3.39 A stakeholder suggested consideration of Adjusted Present Value methodology, based on unlevered beta, as it would be more appropriate in circumstances where project finance is used to finance a single-asset interconnector and debt to equity ratio is constantly shifting.

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Decision and rationale

Financial parameter	Decision
Cost of debt	We have decided to base our approach for calculating rates of return at the floor on a 20-day average of 50% weighting to the iBoxx GBP Non-Financials 15yr+ A rated index (ISIN reference: DE000A0JY894) and a 50% weighting to the iBoxx GBP Non-Financials 15yr+ BBB rated index (ISIN reference: DE000A0JZAP4) and to determine the rates at FID or FC, as appropriate. To convert nominal to real yields we have decided to use CPI inflation target.

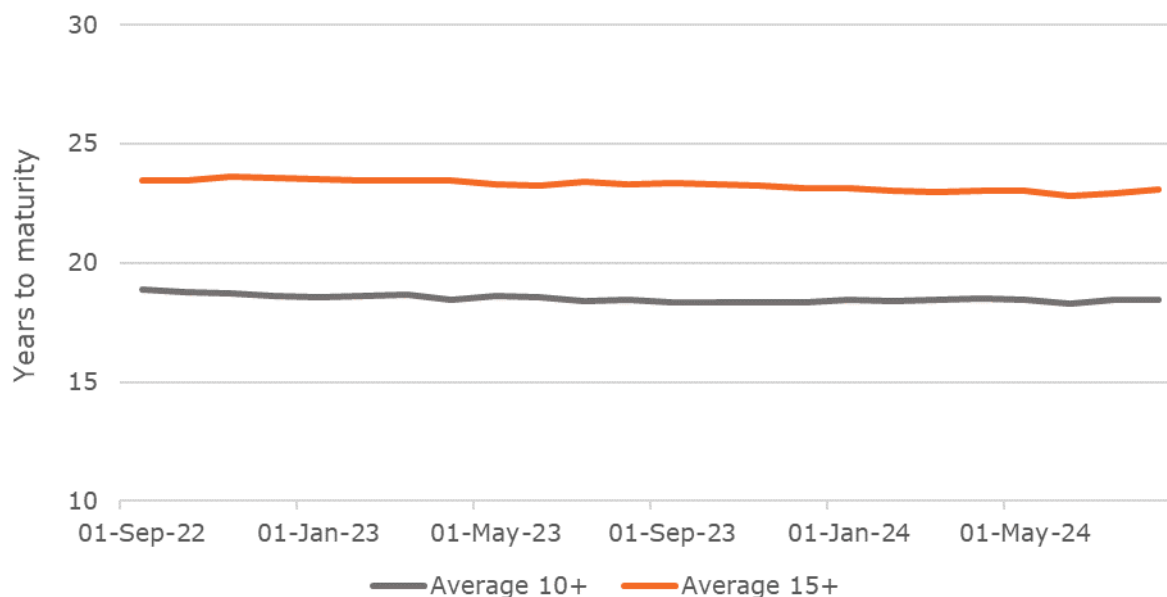
3.40 We have decided to base our approach for calculating rates of return at the floor on a 50% weighting to the iBoxx GBP Non-Financials 15yr+ A rated index and a 50% weighting to the iBoxx GBP Non-Financials 15yr+ BBB rated index. We have decided to deflate the 20-working day trailing average of this measure using the inflation target set by the Government which is currently 2%.¹⁹ These rates will be calculated at FID or FC, as appropriate. Our rationale for following this approach is set out below.

3.41 We consider the move from iBoxx GBP Non-Financials 10yr+ to iBoxx GBP Non-Financials 15yr+ indices reasonable due to the recent observed transactional data as well as analysis of years to maturity of the iBoxx indices with long-term horizon. Analysis of commercially sensitive data indicates that the maturity of debt for interconnection projects tends to more closely align to the iBoxx 15yr+ indices than iBoxx 10yr+.

¹⁹ Bank of England, [Inflation target](#).

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Figure 2: Analysis of years to maturity of iBoxx GBP Non-Financials 10yr+ and 15yr+ indices (A & BBB averages)



Source: IHS Markit

- 3.42 We consider iBoxx Non-Financials indices appropriate for the purpose of estimation cost of debt for interconnectors. In 2018 Ofgem commissioned a report by Cambridge Economic Policy Associates (CEPA) to provide analysis of the component parts of the cost of capital estimates.²⁰ We considered that the evidence presented in this report remains relevant and supports the ongoing use of iBoxx indices. Specifically, the iBoxx Non-Financials corporate indices are broad and calculated using a transparent methodology. We consider that the iBoxx indices are broadly recognised by the market as robust independent sources of debt costs. We regard the methodology, rating qualifying criteria and number of instruments included within the index are likely to derive estimates of debt costs which are reasonably achievable for these assets. We also note use of the iBoxx indices is in line with common UK regulatory practice.
- 3.43 Stakeholders queried how Ofgem intends to deflate the 20-day trailing average from nominal to CPIH-real terms. As noted in paragraph 3.20, we consider it unlikely that long run CPIH inflation expectations would reside below the Bank of England CPI target – therefore, for the purpose of long-term inflation forecast we

²⁰ CEPA (2018), [Review of Cost of Capital Ranges for New Assets for Ofgem's Networks Division](#), pages 9-10.

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will adopt the Bank of England CPI inflation target (2%). We consider this approach to be appropriate for deflating long-term iBoxx indices which are expressed in nominal terms. In line with regulatory convention, we use the Fisher equation²¹ to convert the values from nominal to CPIH-real.

- 3.44 We see benefits in adopting CPIH over CPI, given that it will become the UK headline measure over time. In paragraphs 3.18 and 3.20 we outline how we will seek to address the lack of CPIH forecasts in our approach.
- 3.45 As outlined above, we will not utilise breakeven inflation curves to calculate the long run inflation assumption utilised to convert the allowed return on debt from nominal to real terms, as we are not aware of any CPIH breakeven curves that would be suitable. We also do not consider that utilising RPI with a fixed wedge applied as an approximation for CPIH would offer greater accuracy advantages over our current approach and would increase complexity.
- 3.46 We have not proposed to implement a high and low range. An example of the calculation is included in paragraph 3.51.
- 3.47 We do not consider the proposed approach for Window 3 requires retrospective changes to other determinations. Such an action would conflict with normal regulatory practice. Each regulatory determination is made considering best practice and available evidence at the time and so decisions may reasonably vary with each determination. Retrospective changes may undermine the stability and predictability of the regulatory regime, particularly if a stakeholder(s) regarded the changes as detrimental to their interests.
- 3.48 Some stakeholders requested clarification on the treatment of licensees that adopt a standalone project financed structure. Under the current approach applied to Window 1 and Window 2 projects, developers can apply for the application of two alternative approaches (Variation 4²²):
- Approach 1: To continue using a notional cost of debt approach to set the revenue floor and allow developers to recover both equity and debt investments at the floor. However, we replace the default regime notional cost

²¹ Fisher equation can be expressed as $(1+i)=(1+r)(1+\pi)$, where i equals the nominal interest rate, r equals the real interest rate and π equals the inflation rate.

²² Ofgem (2020), [Decision on proposed changes to our electricity interconnector cap and floor regime](#), page 28.

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of debt benchmark with a single benchmark (UK Non-financial iBoxx BBB 10yr+ years) that better reflects the risk profile of investing in standalone project-financed interconnectors; or

- Approach 2: To set the revenue floor based on a competitive market process overseen by Ofgem. Under this approach, the floor will cover only the debt-gear portion of the investment (including provision for a reasonable debt service cover ratio (DSCR)). However, if the floor based on this approach is higher than the floor based on Approach 1 above, and higher floor payments from consumers are required, developers would reimburse consumers the difference from future revenues above the floor before they can recover equity investment and any dividends.

3.49 For the projects within Window 3 that requested regime variations we have decided to update the Approach 1 presented in the paragraph above to reflect the most recent methodology of deriving the notional cost of debt under the default Window 3 regime. We will set the notional cost of debt benchmark as the UK Non-financial iBoxx BBB 15yr+. Projects will be able to apply for variations under the updated approach 1 or unchanged approach 2 in Window 3.

3.50 An adjusted present value approach, utilising actual company gearing inputs, would violate the 'notional' approach used extensively by regulators when setting allowed returns. We consider that the notional capital structure is an important component of protecting consumers from the risks emanating from companies actual financing decisions. Under this approach, companies can deviate from the notional capital structure assumptions. However, the risks and rewards of these decisions reside with companies and their investors, not consumers.

Illustrative Example

3.51 In Table 1 below, we present an illustrative example of calculated return at the floor. This illustrative example is based on a reference date of 30th September 2024.

Table 1: Return at the floor calculations

Date	Index 1	Index 2	Average (nominal)	Inflation target	Real Average Rate
20 days to FID	A 15yr+	BBB 15yr+	$\text{Avg} = (A + \text{BBB}) / 2$	CPI inflation target	$= (1 + \text{Avg}) / (1 + \text{CPI inflation target}) - 1$
03/09/2024	5.37%	5.91%	5.64%	2.00%	3.57%

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Date	Index 1	Index 2	Average (nominal)	Inflation target	Real Average Rate
04/09/2024	5.34%	5.87%	5.61%	2.00%	3.53%
05/09/2024	5.33%	5.86%	5.60%	2.00%	3.53%
06/09/2024	5.32%	5.84%	5.58%	2.00%	3.51%
09/09/2024	5.34%	5.86%	5.60%	2.00%	3.53%
10/09/2024	5.31%	5.83%	5.57%	2.00%	3.50%
11/09/2024	5.26%	5.79%	5.52%	2.00%	3.46%
12/09/2024	5.29%	5.82%	5.55%	2.00%	3.48%
13/09/2024	5.28%	5.81%	5.54%	2.00%	3.47%
16/09/2024	5.27%	5.80%	5.54%	2.00%	3.47%
17/09/2024	5.27%	5.80%	5.54%	2.00%	3.47%
18/09/2024	5.34%	5.86%	5.60%	2.00%	3.53%
19/09/2024	5.37%	5.89%	5.63%	2.00%	3.56%
20/09/2024	5.39%	5.91%	5.65%	2.00%	3.58%
23/09/2024	5.41%	5.94%	5.67%	2.00%	3.60%
24/09/2024	5.42%	5.94%	5.68%	2.00%	3.61%
25/09/2024	5.46%	5.98%	5.72%	2.00%	3.65%
26/09/2024	5.50%	6.01%	5.75%	2.00%	3.68%
27/09/2024	5.49%	6.00%	5.74%	2.00%	3.67%
30/09/2024	5.49%	6.00%	5.74%	2.00%	3.67%

20-day average to FID (rounded)

3.55%

Source: IHS Markit, BOE

Approach for estimating cap rate for Window 3

Primary cost of equity estimation methodology

3.52 Recommendation 2 of the UKRN Guidance suggests that regulators should continue to use the Capital Asset Pricing Model (CAPM) as their primary approach for estimating the cost of equity.²³

²³ UKRN (2023), [UKRN guidance for regulators on the methodology for setting the cost of capital](#), page 11.

3.53 The CAPM has three inputs, all of which need to be estimated to calculate the cost of equity for interconnectors and set an appropriate allowed return on cap for these projects:

- The Equity Beta (β)
- The Risk-Free Rate (RFR); and
- The Total Market Return (TMR).

3.54 These inputs are combined in the following way to estimate the cost of equity:

$$COE = RFR + \beta * (TMR - RFR)$$

3.55 We discuss our approach to calculating these metrics below.

Estimating Beta (β)

Consultation summary

3.56 The CAPM used to estimate the cost of equity operates under the assumption that specific risks associated with an investment can be diversified away. This implies that investors do not seek additional compensation for exposure to these 'specific' or 'non-systematic' risks. In contrast, the risk that remain is unavoidable or 'systematic' and cannot be diversified and therefore investors require compensation for exposure to this risk. The most commonly referenced systematic risk is exposure to the general performance of the economy.

3.57 Beta is the measure of an asset's exposure to non-diversifiable systematic risk, relative to the average exposure of assets in the market which is set at a beta of 1. Regulators typically estimate beta by calculating the covariance between the price movements of publicly listed company shares and the average price movements of relevant equity indices. This can be done directly for listed companies or indirectly by using listed companies as proxies for unlisted ones.

3.58 The measurement of raw equity beta involves statistical analysis, which can range from basic Ordinary Least Squares (**OLS**) regressions to more advanced techniques like Generalised Autoregressive Conditional Heteroskedasticity (GARCH) models. These methods assess the relationship between asset's returns and the overall market.

- 3.59 Recommendation 5 of the UKRN guidance suggests that regulators should estimate equity beta for the notional capitalised company by using comparable listed companies and applying standard regression techniques (i.e. OLS).²⁴ The guidance further emphasises that if the listed comparator has a different level of gearing than the notional company, regulators should adjust the raw equity beta by 'de-levering' and then 're-levering' it to account for these differences in gearing.
- 3.60 In the Second Consultation we proposed to align our methodology with the UKRN guidance recommendation, stating that we intend to base our beta analysis on OLS regressions of relevant listed comparators. We further explained that we will use de-gearred data to make asset beta comparisons before re-gearing to the notional capital structure in order to estimate the appropriate equity beta input for the CAPM cost of equity.
- 3.61 We proposed to base our calculations on group of comparators, applying equal weightings to each, using daily observations over a 5-year estimation window. In addition, we proposed to apply a notional gearing level of 50% and include a debt beta estimate of 0.075 in the calculations.
- 3.62 Given the absence of pure-play listed comparators for regulated activities such as interconnectors, we considered six listed comparator companies as appropriate inputs into the estimation of an appropriate equity beta for Windows 3 projects. We proposed Drax Power (DRX LN), SSE (SSE LN), RWE (RWE GR), Iberdrola (IBE SM), Orsted (Orsted DC) and National Grid (NG/ LN) as suitable comparators due to their exposure to similar markets and, in aggregate, similar risks and returns to those associated with the interconnector projects.
- 3.63 In the Second Consultation, we asked two questions related to the beta parameter which were as follows:
- Q1. Do you agree with our methodology for calculating equity beta for Window 3 interconnectors? If not, could you please explain why, provide evidence for your reasons and suggest alternatives?

²⁴ UKRN (2023), [UKRN guidance for regulators on the methodology for setting the cost of capital](#), page 25.

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- Q2. Do you agree with the comparators we are proposing to use to calculate the beta parameter? If not, could you please explain why, provide evidence for your reasons and suggest alternative methods?

Decision Overview

Financial parameter	Decision
Asset beta and notional equity beta	We have decided to use an asset beta of 0.60 for the third application window for interconnectors. This figure is derived from an average of the six comparators' asset betas - Drax, SSE, Iberdrola, Orsted, RWE, and National Grid using equal weighting and a 5-year estimation window. We have decided to set the notional equity beta at 1.125, calculated using a 50% notional gearing and a debt beta of 0.075.

3.64 The responses, decisions and rationales are included below and separated by subject matter.

The use of beta comparisons as a methodological framework

Summary of consultation responses

- 3.65 A stakeholder stated that because of no pure play quoted comparables for offshore interconnectors it is difficult to derive an accurate beta from existing listed companies. Another stakeholder said that the proposed listed companies are not directly comparable to interconnectors and therefore not fit for purpose.
- 3.66 Two stakeholders stated that single-asset companies like interconnectors face higher risk leading to greater concentration risk, which should result in a higher beta multiple.
- 3.67 A stakeholder stated that using a project-specific beta might be a more suitable approach rather than beta comparisons.
- 3.68 A stakeholder stated that the beta comparators needed to be more specific to a particular revenue stream of interconnectors such as congestion revenues and the capacity market revenues, which have different risk profiles.
- 3.69 A stakeholder stated that benchmarking using companies that are not solely focused on interconnector activities may require discretionary adjustments to the equity beta, upwards or downwards to account for differences in risk profiles.

Decision and rationale

- 3.70 We have decided to use the beta comparison methodology and have chosen six listed companies for our comparator set. We have chosen Drax Power, SSE, RWE, Iberdrola, Orsted and National Grid and applied equal weighting to each company. We consider that the companies selected, and the methodology applied, provides an aggregated risk profile that aligns with the interconnector cap and floor regime.
- 3.71 There are no pure play listed comparators for regulated activities such as interconnectors. The lack of pure-play options makes it necessary to apply regulatory judgment in selecting appropriate proxies. As a result, we have decided to select a peer group of listed companies that we consider to be, in aggregate, representative of the systematic risk associated with companies operating under the cap and floor regime. The selected companies operate in similarly capital-intensive sectors and face regulatory frameworks that align with the financial and operational risks of interconnectors, providing a suitable basis for beta comparisons.
- 3.72 Our chosen beta methodology is supported by the UKRN Guidance, which notes that the initial step in estimating the equity beta involves identifying publicly listed companies that can serve as reasonable comparators for the regulated activities for which price controls are set. The UKRN further highlights that while the comparator process is straightforward in certain sectors, finding pure-play listed benchmarks can be challenging in practice.²⁵ This is particularly the case in highly specialised sectors such as interconnectors.
- 3.73 We note that the peer-group approach ensures that the beta reflects a comprehensive view of systemic risks by capturing broader market trends and sector-specific dynamics that may not be adequately represented by individual projects or illiquid assets. Our beta methodology is also supported by academic literature such as the work of Bazhutov, D., Betzer, A., & Stehle, R. (2021), in

²⁵ UKRN (2023), [UKRN guidance for regulators on the methodology for setting the cost of capital](#), page 22.

which the authors state that employing a peer group is the only method to estimate the beta of an unlisted firm.²⁶

- 3.74 In addition, we note that CEPA has further observed that pure-play betas are often unobservable, and estimating beta requires a combination of techniques that incorporate both quantitative and qualitative analysis.²⁷ This means that beta estimates typically rely on both comparator beta benchmarking (quantitative analysis) and qualitative analysis of relative risk in relation to benchmarked comparators or other regulatory precedents.
- 3.75 The cap and floor regime, under which interconnectors operate, offers a distinct downside protection that is not available to many of the listed comparators. This regulatory mechanism sets a floor to protect investors from significant losses while also providing a cap on the potential upside gains. As a result, this structure and the impact on the net level of systematic risks faced must also be considered when interpreting beta estimates derived from companies without such regulatory protections.
- 3.76 In relation to the project beta methodology proposed by stakeholders, we specifically note that while in theory a project-beta approach has merit in capturing specific project-related risks, it is also often constrained by a lack of comprehensive data and excessive subjectivity, which can lead to unreliable results compared to more established methodologies.
- 3.77 We recognise that when benchmarking against companies that are not pure plays (i.e., solely cap and floor interconnector businesses), one approach would be to adjust the equity beta to account for differences in risk profiles. However, we consider that such an approach would be susceptible to excessive subjectivity. We consider the diversified operations of the chosen comparators to provide a balanced risk profile that, in aggregate, aligns sufficiently with the risks faced by interconnectors.

²⁶ Forthcoming, Financial Markets and Portfolio Management (2021, Revised 2023), [Beta Estimation in the European Network Regulation context: What Matters, What Doesn't, What is Indispensable](#), page 2.

²⁷ CEPA (2018), [REVIEW OF COST OF CAPITAL RANGES FOR NEW ASSETS FOR OFGEM'S NETWORKS DIVISION](#), page 17.

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The choice of companies used in the beta comparison

Summary of consultation responses

- 3.78 Six stakeholders stated that that National Grid is not a suitable comparator due to the insufficient proportion of its revenue derived from interconnector projects. A stakeholder stated that National Grid shares exhibit bond-like characteristics, given their consistent dividend policies and should not be used for comparison, as the company is primarily a fully regulated network company and, as a result, the systematic risk is significantly different from a standalone interconnector. Another stakeholder stated that National Grid is not an appropriate comparator as it is not a power generation company, which were considered the most appropriate comparable.
- 3.79 Two stakeholders stated that Centrica should be included as a comparator while less emphasis should be placed on Iberdrola.
- 3.80 Two stakeholders stated that BP, Total and Shell should be included in the list of comparators.
- 3.81 A stakeholder stated that SSE, RWE and Orsted should be excluded from the beta comparison as they primarily face only volume risk whereas interconnectors are exposed to both volume and price risk.
- 3.82 Three stakeholders agreed that RWE and Orsted are appropriate comparators and should be among the selected comparators.
- 3.83 Two stakeholders stated that the suggested comparators may include companies in different geographic environments with risks and market conditions that vary greatly from those of single asset interconnectors.
- 3.84 A stakeholder stated that Ofgem has not considered the changes in Drax's business mix, which have led to a decrease in its notional beta.

Decision and rationale

- 3.85 We have decided to retain National Grid, SSE, Drax, RWE, Orsted, and Iberdrola as our comparable companies, while excluding major oil and gas companies, along with Centrica from our peer group.

National Grid is an appropriate comparator

- 3.86 We have selected our set of beta comparators by focusing on companies with exposure to interconnector operations, those that share a risk profile similar to that of the cap and floor regime, and/or those with relevant exposure to

renewable energy. We believe National Grid clearly meets these criteria, making it a relevant comparator. Specifically, National Grid's business aligns with the risk profile of the cap and floor regime and shares exposure to regulated infrastructure.

- 3.87 In support of this assessment, we note that National Grid Ventures (NGV), is a division of National Grid Plc that oversees projects such as electricity interconnectors and other renewable energy projects. NGV operates six interconnectors, including IFA2, NSL, Viking Link, and Nemo Link, which fall under the cap-and-floor regime. This positions NGV as one of the largest developers of interconnectors in Great Britain, providing it with direct exposure to the operational and regulatory risks inherent in this sector.
- 3.88 We also note that National Grid wider portfolio faces regulatory and revenue dynamics similar to those experienced by pure-play interconnector companies. National Grid's beta also reflects its participation in large-scale infrastructure projects and regulated frameworks which provide valuable insights into risk-adjusted returns relevant to long-term infrastructure investors.
- 3.89 The chosen comparators represent a spectrum of systematic risk profiles that we believe interconnectors sits within. It would be inappropriate to consider only comparators at the high-risk end of the spectrum and ignore the lower-risk comparators, as the end result would be skewed, and consumers would be exposed to higher costs than necessary. The purpose of the comparators is to serve as a proxy for the risk profile (beta) of interconnectors in aggregate, rather than addressing them individually.

Iberdrola and Centrica's suitability as comparators

- 3.90 Iberdrola is a major utility company with significant investments in energy infrastructure, including renewable generation (wind, solar, hydro) and transmission networks. Iberdrola's renewable focus complements interconnector operations by facilitating cross-border energy trade, helping balance supply and demand across different markets.
- 3.91 Iberdrola plays a significant role in large-scale transmission projects that align closely with the operational risks and financial risks of standalone interconnector companies. Through its subsidiary, ScottishPower Energy Networks, Iberdrola is a key partner in the £3.4 billion Eastern Link and the Eastern Green Link 1, both key subsea cable projects connecting Scotland to England. These initiatives expose Iberdrola to similar regulatory and operational risks as interconnectors. In

addition, with assets across Europe, the UK, and the US, Iberdrola's experience navigating diverse regulatory frameworks is similar to the cross-border operations of interconnectors, making it a relevant comparator.

- 3.92 Centrica is primarily focused on energy supply, trading, and services after the sale of its upstream oil and gas assets. This difference in business model makes Iberdrola a closer proxy for large-scale infrastructure investment similar to interconnectors.

SSE, RWE and Orsted as relevant comparators

- 3.93 We consider SSE, RWE, and Orsted to be appropriate comparators for interconnectors due to their exposure to a broad range of market risks that are aligned with the risk profile of interconnectors, which face both volume and price risks. These companies, though primarily energy generators, operate in complex markets that expose them to regulatory, operational, and financial challenges similar to those faced by interconnectors.
- 3.94 We note that SSE, with its diverse portfolio including renewable energy, gas-fired power plants, and regulated transmission networks, faces a range of risks across different segments of the energy sector. The company's exposure to fluctuating market conditions and operational risks positions it as a relevant benchmark for interconnectors, which similarly operate in a regulated yet market-sensitive environment.
- 3.95 RWE's is one of the UK's largest power producers, which further justifies its inclusion as a comparator. It has a total installed capacity of 7.3 GW in the UK and 44.4GW globally, which includes renewables like wind and biomass, as well as conventional gas assets. RWE faces both volume and price risks due to its participation in volatile energy markets. This aligns closely with interconnectors, which are faced with not only the volume of energy transmitted but also the pricing dynamics of cross-border electricity flows.
- 3.96 Orsted's extensive operations in the UK offshore wind market, while primarily focused on renewable energy, still face significant regulatory and market risks, especially as the integration of renewables introduces complexities in pricing and market stability. Although Orsted's exposure is primarily volume-driven, its market leadership and involvement in large-scale infrastructure projects mean that it is likely to be exposed to similar policy and market shifts that impact interconnectors.

- 3.97 We disagree with the view that SSE, RWE, and Orsted should be excluded due to their focus on volume risk. This perspective fails to account for the broader risk spectrum these companies face. Their involvement in both regulated and competitive markets, coupled with their operational scale and the complexities of integrating renewable energy, aligns their risk profile more closely with that of interconnectors than a pure volume-risk perspective would suggest.

Oil and gas companies as appropriate comparators

- 3.98 We have evaluated alternative or additional comparators, including major energy companies such as BP, Shell and Total, and have concluded that they are not appropriate comparators.
- 3.99 We note that while both interconnectors and offshore oil drilling share some operational challenges, particularly in marine environments, their risk exposures and business focus are fundamentally different. Interconnectors focus on laying transmission cables to connect electrical grids across borders, with minimal drilling and lower long-term risks once installed. In contrast, offshore oil drilling is a far more complex operation, requiring extensive drilling through multiple geological layers, often involving high environmental and operational risks.
- 3.100 We consider that trading and hedging activities of the oil majors do not align with those of interconnectors. BP and Shell engage in spot and futures trading in oil, gas, and refined products, which are highly volatile and driven by global supply-demand dynamics, geopolitical events, and storage constraints. Futures contracts allow them to hedge against price fluctuations and manage the risks inherent in volatile energy markets.
- 3.101 Interconnectors, on the other hand, do not directly participate in trading or hedging activities. Instead, their role is limited to operating the infrastructure and facilitating electricity trading between countries. Electricity trading over interconnectors is conducted by third party traders who secure capacity rights through capacity auctions. Interconnectors revenues are derived from facilitating these trading activities. Revenues for interconnectors are influenced by electricity market regulations, transmission constraints, and renewable energy integration rather than speculative trading and global price volatility seen in oil and gas markets.

Geographic and regulatory variance in comparators

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- 3.102 We acknowledge that the proposed comparators operate in varying geographic regulatory environments, which can lead to differences in risk and market conditions compared to single-asset interconnector projects.
- 3.103 While some stakeholders opposed the inclusion of Iberdrola as a comparator due to its operations outside the UK, we note that Orsted and RWE also have significant non-UK operations yet are still accepted by some stakeholders as suitable comparators. We emphasise that comparator selection must avoid bias or cherry-picking of higher beta figures, but instead reflect a balanced approach, considering both operational scale and risk profiles.
- 3.104 We consider that large, diversified energy companies are likely to face similar overarching risks that transcend local regulatory environments. These firms often invest in multiple infrastructure projects across different jurisdictions, thereby gaining a comprehensive understanding of the regulatory landscapes and market dynamics.
- 3.105 We also note that interconnectors, by their nature, are designed to facilitate cross-border electricity trading, which inherently involves exposure to multiple regulatory regimes and market conditions.

Consideration of Drax's business mix

- 3.106 We previously highlighted that the business changes at Drax had made it potentially less suitable as a sole comparator for interconnector beta calculations.
- 3.107 The change in Drax's mix of operations, particularly its transition from coal to biomass and its increasing focus on renewable energy generation, has altered its exposure to market risks, making it less purely comparable to the regulated environment in which interconnectors operate. In recognition of this, Drax's weighting in the beta calculation was reduced from 100% to 16.67%.

Single asset risk in project financed interconnectors

Summary of consultation responses

- 3.108 A stakeholder stated that the stand-alone interconnectors face project risks such as regulatory approvals, delays, and operational challenges, warranting a tailored approach to setting the cap and floor parameters due to this concentration risk.
- 3.109 A stakeholder stated that the proposed comparator companies have diversified business and revenue streams, allowing access to capital markets with relative ease and reduce their cost of capital.

- 3.110 A stakeholder stated that interconnector projects operate in more volatile environments with revenues subject to fluctuations in electricity prices. This revenue volatility makes single-asset projects riskier compared to firms with regulated, stable income. The developer further also pointed out that other regulatory frameworks can offer mechanisms that mitigate market risk, such as cost pass-throughs, rate adjustments, or capacity payments.
- 3.111 A stakeholder stated that investors in large, diversified companies may accept lower returns due to the lower perceived risk and more stable cash flows.

Decision and Rationale

- 3.112 We do not consider it appropriate to make specific adjustments for single asset risks. We consider this risk to be largely diversifiable for investors in this sector. Providing compensation for a diversifiable risk is not required within the CAPM framework and would likely lead to the overcompensation of investors at the expense of consumers.
- 3.113 We note that while large energy companies have diversified portfolios and established operational frameworks, they still face risks comparable to those of interconnectors. Despite their scale, these firms are heavily invested in long-term, capital-intensive projects that are exposed to regulatory, operational, technological, and market uncertainties. Large-scale transmission and renewable energy projects frequently encounter delays, cost overruns, and shifting regulatory landscapes, which can exert financial strain. Moreover, the revenue volatility and potential disruptions seen in interconnectors are similarly present in these companies' major infrastructure projects, where underperformance can be particularly impactful due to the high fixed costs and capital requirements involved.
- 3.114 Interconnectors operate under regulatory regimes that offer significant risk mitigation. Interconnectors, particularly those under frameworks like the Cap and Floor regime, benefit from regulatory protections, including mechanisms that address revenue volatility. These protections include pass-through allowances for increases in operating expenditures (OPEX), adjustments for capital expenditure (CAPEX) cost overruns, and inflation-linked revenue adjustments. These features effectively cushion interconnectors from many of the risks caused by fluctuating electricity prices, demand, and cross-border trading conditions.
- 3.115 The risks associated with concentrated revenue streams and high leverage in single-asset projects do not automatically translate to a linear relationship with

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beta. The regulatory framework for interconnectors, which involves regulated revenue mechanisms, provides a level of downside protection that can mitigate some of the risks that might be associated with single-asset developments.

3.116 As noted in paragraph 3.71 we do not have pure-play listed comparators on which to base our beta estimate. To address this, we have created a cohort of comparators that we consider, in aggregate, to provide a reflection of the exposure to systematic risks of companies participating in Window 3. While these companies may not directly reflect 'single asset risks', neither do the entirety of their operations benefit from the significant investor protections provided by the cap and floor model. Please see paragraphs 3.78-3.107 for further discussion of our assessment of the appropriateness of potential comparators.

Estimation window and data frequency

Summary of consultation responses

3.117 A stakeholder suggested including the 2-year beta estimate in addition to the 5-year beta currently being considered.

Decision and Rationale

3.118 We have decided to adopt a 5-year estimation window, based on simple average of daily returns data, as the basis for estimating beta for interconnectors.

3.119 There are potential trade-offs to consider when setting an estimation window. For example, a 2-year estimation window may be most reflective of current market dynamics. However, it may also lack the breath of datapoints required to accurately assess risk through different market cycles. Such short-term windows are more susceptible to volatility and may not yield stable estimates under varying economic conditions. We consider a 5-year estimation period to provide a more comprehensive view of risk exposure over time whilst remaining sufficiently contemporary and reflective of likely future systematic risk exposure.

The application of debt beta and notional gearing

Summary of consultation responses

3.120 A stakeholder stated that debt beta may not be appropriate for the regime, as the guaranteed floor serves as a form of credit support, effectively reducing the risk to debt holders to a level comparable to sovereign debt and thus the debt-beta should be set at zero.

3.121 Two stakeholders noted that the notional gearing in the RIIO-3 (60%) is higher than in interconnector cap and floor (50%), which impacts the cost of equity. As a

result, they suggested that the level of assumed gearing for interconnectors should be reassessed. A stakeholder further argued that there is a correlation between debt beta and notional gearing and since notional gearing for RIIO is higher than for the cap and floor regime while debt betas for each regime are the same, then debt beta for the cap and floor regime should be removed until further consideration is given on this matter.

- 3.122 A stakeholder stated that the proposed comparators include companies with lower leverage, diversified financing options, and a mix of debt and equity across their operations. They also mentioned that these companies can access capital markets with relative ease, reducing their cost of capital.

Decision and rationale

- 3.123 We have decided to set the debt beta at 0.075 with a notional gearing of 50% to accurately reflect the risk profile of interconnectors, while accounting for operational and market-related risks.
- 3.124 We note that the cap-and-floor regulation offers a guaranteed floor return for interconnectors, suggesting a low-risk profile. However, the cost of debt assessment at A/BBB credit ratings demonstrates that investors do not consider lending to interconnectors to be risk-free or otherwise comparable to sovereign debt.
- 3.125 We note that the UKRN listed debt betas used by other regulators are in the range of 0.05 to 0.125.²⁸ This range reflects the correlation of a utility's debt with broader market movements, considering the specific risks and characteristics of the regulated assets.
- 3.126 Regulatory precedent shows that similar debt beta estimates have been used by regulators when setting allowed returns, suggesting that debt beta assumptions do not vary significantly within the 50-60% notional gearing levels used by major regulators. As an example, Ofgem uses 0.075 as the debt beta estimate for the purpose of the 'early view' of the cost of equity in the RIIO-3 Sector Specific Methodology Decision (SSMD)²⁹ at both 55% and 60% notional gearing, Ofwat

²⁸ UKRN (2023), [Cost of Capital – Annual Update Report](#), page 10.

²⁹ Ofgem (2024), [Decision –RIIO-3 Sector Specific Methodology Decision – Finance Annex](#), pages 93-94.

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uses range of 0.05-0.15 at 55% gearing,³⁰ and the CMA recently used a debt beta midpoint estimate of 0.075 at 60% gearing.³¹

3.127 Our assessment of an appropriate notional capital structure is consistent with Window 2 and is discussed further in the Second Consultation. As we have not received any material evidence that this should be changed, we consider that 50% gearing reflects an appropriate capital structure for this activity and is not directly influenced by the comparators used to estimate beta.

General comments

Summary of consultation responses

3.128 A stakeholder argued that Ofgem has presented no evidence to suggest that there is a need to review and amend the equity beta for Window 3 interconnector projects, in comparison to what has been utilised in Window 1 and Window 2.

3.129 A stakeholder observed a potential inconsistency in Ofgem's approach to calculating the gearing of RWE within the comparator set.

Decision and rationale

3.130 We consider that updating our approaches and calculations to reflect best practice is a standard element of regulatory regimes and a requirement to meet our duties. We have explained above both the reasons for reducing reliance on Drax as a comparator and the detailed rationale for our selection of comparators.

3.131 In the consultation, we adjusted the Bloomberg data on the RWE's net debt position to account for substantial provisions in their financial statements. After considering the stakeholder's evidence, we have decided to adopt a consistent methodology based on Bloomberg data when calculating net debt and enterprise value for all comparators. There are likely to be both benefits and costs of making manual adjustments to data that is widely utilised by financial market participants (i.e., from a reputable source such as Bloomberg) – with the potential for increased accuracy offset by risks such as subjective or inconsistent adjustments across the data set. On balance, we consider that the decision to use unadjusted data for all comparators is appropriate and helps to ensure that our approach is as consistent and transparent as possible.

³⁰ Ofwat (2023), [PR24 Final Methodology](#), page 47.

³¹ CMA (2021), PR19 Redetermination Final Report, page 881.

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Final beta calculations

3.132 In the Table 2 below we present our estimate of the appropriate asset beta for Window 3 of 0.60. The final calculation is based on data as of the reference date of 30 September 2024.

3.133 The asset beta value is estimated based on the asset betas calculated for six comparators: Drax Power (DRX LN), SSE (SSE LN), RWE (RWE GR), Iberdrola (IBE SM), Orsted (Orsted DC) and National Grid (NG/ LN). We place equal weight on each comparator (approximately 16.67%, representing, 100% divided by 6 comparators).

3.134 Raw equity betas for each comparator are 'de-gearred' (based on the average net debt to enterprise value over 5-year period) to strip out the impact of the level of debt within the capital structure of each firm (assuming a zero-debt beta) to find an unlevered beta.

3.135 This unlevered beta for each comparator is combined with our 0.075 debt beta assumption to allow us to compare the asset betas of relevant comparators.

3.136 The analysis uses FTSE All-Share index data for the UK comparators which are Drax, SSE and National Grid and the STOXX Europe 600 index data for RWE, Iberdrola and Orsted.

Table 2: Asset beta calculations for each comparator

Component	DRX LN	SSE LN	RWE GR	IBE SM	Orsted DC	NG/LN	Ref	Equation
Spot raw beta	0.915	0.971	0.863	0.754	0.717	0.608	A	
Observed gearing over period	37.64%	32.22%	5.85%	36.14%	9.99%	50.11%	B	
Unlevered beta	0.570	0.658	0.813	0.482	0.645	0.303	C	$=A*(1-B)$
Debt beta	0.075	0.075	0.075	0.075	0.075	0.075	D	
Asset beta	0.599	0.682	0.817	0.509	0.652	0.341	E	$=C+(B*D)$
Weighting	16.67%	16.67%	16.67%	16.67%	16.67%	16.67%		

Weighted average of comparator companies' asset betas: 0.600

Source: Bloomberg, companies' financial statements, Ofgem's analysis

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3.137 The average asset beta of our comparator set is 're-gear'd' in line with the notional capital structure, which is 50% gearing for Window 3 projects. This results in an equity beta for projects within Window 3 of 1.125 which is shown in Table 3. The equity beta parameter is a required input to the CAPM when estimating cost of equity for Window 3 projects.

Table 3: Equity beta for Interconnectors

Component	Value	Ref	Equation
Weighted average of companies' asset betas	0.60	A	
Notional gearing	50%	B	
Debt beta	0.075	C	
Equity beta for Window 3 Interconnectors	1.125	D	$D = [A - (B \times C)] / (1 - B)$

Estimating Risk-free rate (RFR)

Consultation summary

3.138 The RFR is, in theory, the rate of return required to invest at zero risk. In practice, no investment is truly risk-free, so this hypothetical risk-free rate of return must be estimated.

3.139 We consulted on using 20-day simple trailing average of the 20-year index-linked gilts (ILGs), expressed in CPIH-real terms. As the RFR is a market-wide rather than asset specific parameter, we proposed that the methodology of selecting the instruments used to estimating RFR as well as converting rates to CPIH-real terms should be aligned with the methodology following the Ofgem's latest thinking regarding RIIO price controls for gas and electricity transmission networks as updated from time to time. We stated that RFR rates will be calculated at FID or FC, as appropriate.

3.140 We asked stakeholders if they agree with the issues raised and the proposed changes to the cost of equity.

Summary of consultation responses

3.141 A stakeholder stated there is some logic to achieve consistency across market parameters such as the risk-free rate. Another stakeholder stated that the proposed approach to RFR aligns to latest thinking across UK regulators and

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allows the parameter to be set near to financial close, reducing the risk of divergence between Ofgem’s view and the actual RFR at the time. They added that the proposed approach of using Indexed-Linked Gilts, appropriately converted to CPIH terms appears a better approach than a fixed value set far ahead of financial close.

3.142 A stakeholder stated that to estimate the RFR, Ofgem needs to take account of the “convenience premium” in government bonds. They also stated that using the 20-day average of the 20-year Index-Linked Gilts may not be appropriate and a “forward curve” adjustment should be included in calculations.

3.143 A stakeholder asked for more clarity on how RFR will be calculated, particularly on how Ofgem intends to convert an observed RPI real yield to a CPIH-real rate.

3.144 A stakeholder stated that the current proposals for indexation of the RFR and the determination of Equity Risk Premium (ERP) produce a counterintuitive result of declining returns when the risk-free rate rises. They also stated that where there is uncertainty on the methodology applied, such as the impact on the ERP of indexation of the RFR, the cap and floor regime needs to remain unchanged.

3.145 A stakeholder noted that RIIO may not be an appropriate analogue as it is regularly reviewed, while the Cap and Floor is fixed for 25 years.

Decision and rationale

Financial parameter	Decision
Risk-free rate (RFR)	We have decided to calculate the RFR using the 20-day average of 20-year Index Linked Gilts (ILG) yield at FID or FC, as appropriate. To convert RPI-real to CPIH-real yields we have decided to apply RPI-CPI wedge.

3.146 We have decided to set the RFR based on the 20-day average of 20-year ILG yield. The RFR will be set for each project at its FID or FC (as appropriate) and will apply to the whole operational period of the project. We do not intend to ‘index’ the RFR over time, as suggested in some stakeholder responses to the consultation. We have decided not to include other proxies or adjustments in our

estimate which aligns with Ofgem’s latest methodology decisions for the RIIO price controls.³² Our rationale for following this approach is set out below.

- 3.147 Regarding the concern raised by the stakeholder that RIIO may not be an appropriate benchmark for the interconnector regime as the RIIO controls are regularly reviewed, we note that the cost of capital set in RIIO price controls reflects Ofgem’s estimate of the long-term cost of capital at a particular level of risk. This is already in line with the approach we deploy within the interconnector regimes when setting a long-term real cost of capital at FID or FC (as appropriate). By utilising the regular RIIO price control analysis of the latest evidence and regulatory best practice in relation to the market-wide parameters (RFR and TMR), we can ensure that the cost of capital within the interconnector regime is as accurate as possible. We also see benefits to stakeholders from an increase in the consistency of approach taken across Ofgem and other relevant regulators and regimes.
- 3.148 We consider using ILGs as the sole proxy for the RFR to be appropriate, noting that it is the approach used by both Ofgem and Ofwat when setting price controls.^{33 34} This approach is also in line with UKRN Guidance Recommendation 3 - that to estimate the RFR within the CAPM, regulators should use recent yields on the index-linked gilts, with a maturity which matches the assumed investment horizon for their sector. The UKRN Guidance also notes that, consistent with recent precedent, maturities of 10 to 20 years are likely to be suitable for most sectors.³⁵ We consider it appropriate to use the instruments with a maturity of 20 years for the cap and floor interconnector regime for which operational period is 25 years.
- 3.149 We do not intend to include other proxies or adjustments in our estimate as we do not consider these to be likely to increase the accuracy of our estimate. The rationale for this view is discussed further in the paragraphs 3.150-3.153 below.

³² Ofgem (2024), [Decision –RIIO-3 Sector Specific Methodology Decision – Finance Annex](#).

³³ Ofgem (2024), [Decision –RIIO-3 Sector Specific Methodology Decision – Finance Annex](#), pages 52-53.

³⁴ Ofwat (2023), [PR24 Final Methodology](#), page 15.

³⁵ UKRN (2023), [UKRN guidance for regulators on the methodology for setting the cost of capital](#), page 14.

- 3.150 A stakeholder stated that to estimate the RFR, Ofgem needs to take account of the “convenience premium” in government bonds. However, as was recently highlighted by the taskforce behind the UKRN Guidance, we are aware of no empirical estimates of the convenience yield in ILGs at the 10-20 year CAPM investment horizon used by most regulators.³⁶ The UKRN refers to the academic paper with estimates of the convenience yield in a UK context in which Diamond and Van Tassel (2023) note a key driver of the convenience yield is the money-like feature of government treasuries.³⁷ In our assessment it is likely that money-like features are strongest in short-dated government instruments and weakest in longer-dated government instruments. It would seem illogical to use longer-dated instruments, with significantly higher duration risk, for money-like purposes.
- 3.151 We also note that Diamond and Van Tassel highlight the correlation between convenience yields and periods of market dislocation: 'convenience yields...spike during the 2007-2009 US financial crisis, 2011-2012 Euro crisis and 2016-2017 Brexit crisis. Convenience yields grow the most in the country where each crisis is centered'.³⁸ In line with the analysis presented in the SSMD, we interpret this observation as suggesting that the value of the 'convenience yield' does not necessarily correlate with the conveniences it is argued to represent (i.e., changes in the convenience of using government bonds as 'money' over time). An alternative interpretation of this observation is that in times of financial distress, non-government assets are not considered to be risk free (while in times of relative calm, when investors have high confidence in the future, a range of assets might approach the yields on assets considered to be genuinely risk free). This suggests to us that, on the basis that future market conditions are unknowable, ILG yields are a significantly better proxy for estimating the RFR than other instruments, including those with a higher credit rating than the UK government, such as AAA non-government bonds.

³⁶ UKRN (2023), [UKRN guidance for regulators on the methodology for setting the cost of capital](#), page 14.

³⁷ Diamond and Van Tassel (2023), [Risk free rates and convenience yields around the world](#).

³⁸ Diamond and Van Tassel (2023), [Risk free rates and convenience yields around the world](#), page 4.

- 3.152 We note that Ofgem's use of ILG's as the sole proxy for the RFR was previously considered by the CMA during the appeal of RIIO-2 and was considered to be 'not wrong'.³⁹
- 3.153 A stakeholder's comment suggested that our estimate of the RFR should include a "forward curve" adjustment. We do not consider it appropriate to adjust our RFR estimate to take account of implied forward rates. As discussed by the CMA in the Redetermination of PR19,⁴⁰ such 'forward rate adjustments' do not seem to provide a more accurate estimate of future spot rates than current spot rates and so are likely to impair rather than improve our estimate of the RFR.
- 3.154 Regulatory judgement must be used when deciding how much data to include in an estimate of the RFR, and specifically what the length of the trailing average should be. We have decided to base our calculation on 20-day working days average to the date of the FID of each project. A shorter averaging period may suggest the market's latest thinking on an RFR that is likely to be appropriate for the period in question but may also be subject to greater short-term volatility in reaction to recent market events. Conversely, a longer averaging period may help to smooth out short-term volatility but may contain historical information on market rates that are no longer appropriate on a forward-looking basis. On balance, we consider a 20-day averaging period is sufficient to avoid capturing particularly volatile pricing points whilst allowing our estimate to be responsive to current market conditions and expectations. We note that this is the approach used by Ofwat when estimating the (unindexed) RFR within its price control decisions, as well as being consistent with our approach for calculating the rate of return at the floor.
- 3.155 Regarding the caution expressed on the lack of clarity on how Ofgem intends to convert an observed RPI real yield to a CPIH-real rate, we acknowledge this concern. ILGs are 'RPI-real' instruments - they do not pay an annual coupon, but their value is uplifted annually by outturn RPI inflation. RPI inflation is considered by the OBR to be approximately 90bps higher than CPI inflation due to the

³⁹ CMA (2021), [Network Companies vs GEMA, Final Determination, Volume 2A: Joined Grounds: Cost of equity](#), paragraph 5.107.

⁴⁰ CMA (2021), [PR19 Redetermination – Final Report, paragraphs](#), 9.228 - 9.234

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method of calculation.⁴¹ To use ILG yields as a proxy for the RFR, we must adjust yields to 'CPIH-real' terms by estimating the difference between future CPIH and RPI inflation - often referred to as the inflation 'wedge'.

3.156 We have decided to adjust ILG yields to CPIH-real terms on the basis of a 'wedge' calculated primarily using the official forecasts methodology. We do not consider there to be sufficient evidence to justify further adjustments for differences between CPI and CPIH.

3.157 When calculating the forward looking CPIH-RPI wedge we considered the anticipated conversion of CPIH and RPI inflation methodologies from February 2030. To address this, we have calculated the wedge using the approach shown in Table 4 below.

Table 4: RPI-CPIH wedge forecast

Year	RPI forecast	CPIH forecast	% of 'legacy' RPI in RPI figure
2024/25	2.4%	1.6%	100%
2025/26	2.2%	1.6%	100%
2026/27	2.6%	1.7%	100%
2027/28	3.0%	2.0%	100%
2028/29	2.9%	2.0%	100%
2029/30	2.83%	2.0%	92%
2030/31	2.38%	2.0%	42%
2031/32-2043/44	2.0%	2.0%	0%

Source: Ofgem analysis of OBR data

3.158 We have based our calculation on the 20-year geometric average wedge required over the 20-year tenor of our RFR proxy. To do this, we take medium term OBR forecasts for RPI and CPI for each financial year that are available.⁴² For 2029/2030, we take 92% of a long run RPI assumption (we assume 2.9% based on a 0.90% forward looking CPI-RPI wedge) and 8% of a long run CPIH

⁴¹ OBR (2019), [Forecast evaluation report](#), Box, 2.3: Long run wedge between RPI and CPI inflation

⁴² We use data from Economic and Fiscal Outlook, Detailed Forecast Tables Economy, March 2024 edition.

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assumption (assume 2%) to generate an RPI forecast of 2.83%.⁴³ For 2030/31 we take 42% of the long run RPI assumption and 58% of the long run CPIH assumption to obtain an RPI forecast of 2.38%.⁴⁴ For financial years 2031/32 and beyond we assume both RPI and CPIH are equal to 2% and the wedge is effectively zero. We continue to use CPI forecasts as a proxy for CPIH expectations.

Illustrative example

3.159 In the table below, we present an illustrative example of RFR parameter. This illustrative example is based on a reference date of 30th September 2024.

3.160 ILG Yield were calculated based on 20-day average of Government Liability Curve (GLC) datapoints and expressed in RPI-real terms as shown in Table 5.

Table 5: ILGs Yield

Date	Index	Date	Index
20 days to FID	GLC real	20 days to FID	GLC real
03/09/2024	1.18%	17/09/2024	1.04%
04/09/2024	1.15%	18/09/2024	1.04%
05/09/2024	1.13%	19/09/2024	1.09%
06/09/2024	1.13%	20/09/2024	1.12%
09/09/2024	1.12%	23/09/2024	1.13%
10/09/2024	1.13%	24/09/2024	1.13%
11/09/2024	1.10%	25/09/2024	1.18%
12/09/2024	1.11%	26/09/2024	1.20%
13/09/2024	1.08%	27/09/2024	1.21%
16/09/2024	1.06%	30/09/2024	1.21%

20-day average to FID (rounded): 1.13%

Source: Bank of England, real GLC (spot)

3.161 We have calculated illustrative RFR as a sum of ILG Yield and RPI-CPIH wedge.

Table 6: Summary of RFR calculations

⁴³ This assumes for April 2028 to March 2030 (all months used in the calculation of the FY average change), twenty-two out of the twenty-four months are using the old RPI definition. The RPI definition is expected to align with CPIH in February 2030.

⁴⁴ As above, this assumes ten out of the twenty-four months between April 2029 and March 2031 use the old RPI definition.

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Metric	Value
ILG Yield	1.13%
RPI-CPIH wedge (2024/25)	0.26%
RFR (illustrative value on 30 September 2024)	1.39%

Source: Ofgem’s analysis of Bank of England and OBR data

Estimating Total Market Return (TMR)

Consultation summary

- 3.162 The TMR is an estimate of the return that investors expect for taking the market average level of risk. The TMR is an estimate and cannot be definitively calculated in advance.
- 3.163 As the TMR is a market-wide rather than asset specific parameter, in the Second Consultation we proposed that the TMR range is aligned with the range estimated using the Ofgem’s latest methodology used in the RIIO price controls for gas and electricity transmission networks as updated from time to time.
- 3.164 We asked stakeholders if they agree with the issues raised and the proposed changes to the cost of equity.

Summary of consultation responses

- 3.165 A stakeholder stated that the proposed approach for determining the TMR parameter does not sufficiently account for the higher risk environment that offshore interconnectors operate in compared to traditional onshore assets. They expressed concern that the proposed TMR methodology relies heavily on historical data which may not effectively capture the current and future landscape of offshore interconnectors. The stakeholder suggested that adopting a forward-looking approach for TMR that incorporates recent market data and trends would better reflect the current risk environment.
- 3.166 A stakeholder stated that the proposed TMR framework for regulated business including RIIO is not appropriate and should instead be derived using equity returns of renewable RO and independent power generation, which is more likely to require similar returns as stand-alone interconnectors.
- 3.167 A stakeholder recommended that Ofgem increase the TMR estimation to ensure the gap between the cost of equity and cost of debt remains commensurate with the risk exposure for both equity and debt investors.

- 3.168 A stakeholder stated that the use of RIIO-2 methodology is outdated, due to the development of RIIO-3 in relation to the TMR.
- 3.169 Other stakeholder stated that Ofgem has provided little justification for the proposed amendments to the TMR parameter, other than alignment with RIIO-2. They expressed view that there is no clear rationale or evidence from moving away from the approach used within the first and the second application windows when TMR was set using the latest available value of arithmetic mean of the UK real equity returns from Dimson, Marsh and Staunton (DMS) for the data series starting in 1900.
- 3.170 A stakeholder stated that the TMR parameter should stay aligned with DMS or Ofgem should consider increasing its currently proposed RIIO-3 range or aim at the top of RIIO-3 range. They expressed concern that an alignment to the RIIO-3 SSMD TMR mid-point will increase the risk that TMR is too low. In support of this view, the stakeholder referred to an Oxera report⁴⁵ (prepared for the ENA and submitted as part of the RIIO-3 consultation) stating a TMR range between 6.5% and 7.5% (CPIH-real) would be appropriate and a Frontier Economics report⁴⁶ (prepared for National Grid Electricity Transmission and submitted as part of the RIIO-3 consultation) which stated that a TMR range between 7.55% and 7.86% (CPIH-real) should be considered as a crosscheck and would support a TMR of 7%+ (CPIH-real).
- 3.171 The same stakeholder stated that in the context of interconnectors, a stable TMR and equity beta greater than one results in a perverse outcome where the cap falls as interest rates rise, while the floor would increase. They also noted that Ofgem's latest methodology for estimating TMR, which is RIIO-3 SSMD, is based on the range of parameters and stated that Ofgem should confirm the approach of selecting a point value within that range.

⁴⁵ Oxera (2024), RIIO-3 Cost of Equity, Prepared for Energy Market Association.

⁴⁶ Frontier Economics (2024), The relationship between total market return and gilt yields, Prepared for National Grid Electricity Transmission.

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Decision and rationale

Financial parameter	Decision
Total Market Return (TMR)	We have decided to align the TMR with the range estimated using the Ofgem’s latest thinking regarding RIIO price controls for gas and electricity transmission networks, as updated periodically and to use the midpoint of this range.

3.172 We have decided to align the TMR parameter for interconnectors within Window 3 with the range estimated using Ofgem’s latest methodology for the RIIO price controls at the time of the FID or FC (as appropriate) for each project and to use the midpoint of this range.

3.173 Ofgem’s most recent methodology of estimating TMR is described in RIIO-3 SSMD⁴⁷ where we have decided to set a CPIH-real TMR range of 6.5% to 7.0% based on ex-ante and ex-post methodologies. We continue base our estimates on equity return only, in line with the approach used by market participants and all regulators in the UK. We do not intend to base our methodology on RIIO-2 price controls as raised by one of the stakeholders.

3.174 To estimate the ex-post TMR parameter that defines the top of our TMR range, we base our analysis on the arithmetic average of the 2024 Dimson, Marsh and Staunton (DMS) dataset and inflation dataset which is a combination of the CED (1900 -1947), ONS's CPIH backcast (1950 - 1988) and ONS 'actual' CPIH datasets.

3.175 To estimate the ex-ante TMR parameter we have decided to base our analysis on a version of the ‘decompositional’ approach deployed by the CMA in the redetermination of PR19.⁴⁸ This approach uses inflation adjusted 2024 DMS data on historical average dividend yield and adds this to the historical average of dividend growth in order to estimate an ex-ante TMR based on the key building blocks of investor return expectations.

3.176 Further details of our analysis approach can be found in the RIIO-3 methodology decision. We note that this approach is in line with the UKRN guidance, which

⁴⁷ Ofgem (2024), [Decision –RIIO-3 Sector Specific Methodology Decision – Finance Annex](#), page 68.

⁴⁸ CMA (2021), [PR19 Redetermination – Final Report](#).

recommends that the TMR should be primarily based on historical ex-post and historical ex-ante evidence.

- 3.177 We disagree with the view that TMR parameter does not sufficiently account for the higher risk environment in which offshore interconnectors operate. The TMR metric within the CAPM measures the expected return for an investor holding a diversified portfolio of securities and is not specific to the asset being assessed. The beta metric is the only element of the CAPM that should be tailored to the risks and characteristics of the asset being assessed.
- 3.178 Regarding the suggestion for Ofgem to increase the TMR estimation to ensure the gap between the cost of equity and cost of debt remains commensurate with the risk exposure for both equity and debt investors, we note that the TMR is an estimate of long-term investor expectation of return. As noted in the UKRN Guidance, long-run equity returns are relatively stable over time. In the short term, relative equity and debt premiums can expand and contract. As we are estimating the long-term return required for long-life infrastructure, we continue to consider the stable TMR approach and the use of very long-term returns data to be the most appropriate estimate methodology. For the same reason, we do not think it is necessary to adjust our estimate for current short-term forward looking expectations.
- 3.179 We do not think that DMS approach or selecting the higher end of the price controls range is a reasonable approach and consider both the ex-post and ex-ante approaches to be valid evidence of long-term investor return expectations.
- 3.180 A stakeholder has raised the potential issue of incompatibility between the stable TMR approach and sectors with a beta greater than 1. We note that this situation is not unique to Ofgem or interconnectors, with the CAA setting price controls for Heathrow that also include a stable TMR and a beta greater than one.⁴⁹

Illustrative example

- 3.181 In RIIO-3 SSMD we have decided to present this range as 6.50% (based on the ex-ante analysis) to 7.00% (based on the ex-post analysis) which would result in a midpoint of 6.75% used for an interconnector project if project reaches FID on 30th September 2024. Please note that our decision is to set the TMR in line with

⁴⁹ CAA (2022), [Economic regulation of Heathrow Airport Limited: H7 Final Proposals, Section 3: Financial issues and implementation](#), paragraph 9.195

the latest 'RIIO' approach, and so any new evidence on this matter or updates to best practice will be captured through this process.

Equity transaction cost

Consultation summary

3.182 The cap and floor regime has a transaction cost allowance for raising equity finance. In the default regime, and the regime option for project finance Approach 1, this allowance is calculated by applying 5% to the notional equity for the construction period. In the case of projects availing of the regime option for project finance Approach 2, actual equity transaction cost is applicable.

3.183 The First Consultation sought views from stakeholders on whether it would be appropriate to use a lower equity transaction cost. To inform our decision, we stated that we would welcome evidence on actual transaction costs from stakeholders. We referred to a report⁵⁰ for Ofgem by CEPA in 2018 which suggested allowing up to 3% of the required equity amount for transaction costs.

Summary of consultation responses

3.184 Non-confidential responses on this topic were received from five stakeholders, all developers (or in the same group as a developer) and thus having leading roles in sourcing or providing equity for interconnector projects. Three of the stakeholders have interconnector projects in Window 3 and one has interconnector projects with licences under the cap and floor regime from previous application windows. All of the respondents disagreed with the recommendation in CEPA's report⁵⁰ of 2018 of an equity transaction cost allowance of up to 3%.

3.185 The responses took four broad approaches, sometimes in combination:

- Referring to costs such as underwriting costs and forms of return such as development fees as a basis, or part of a basis, for justification. Investor due diligence costs, legal fees and accounting costs were also cited.

⁵⁰ CEPA (2018), Review of Cost of Capital Ranges for New Assets for OFGEM's Networks Division – Final Report [cepareport_newassets_july2018_final.pdf \(ofgem.gov.uk\)](https://www.ofgem.gov.uk/consult/condocs/cepareport_newassets_july2018_final.pdf)

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- Pointing out the age of the CEPA report, and that it in turn had referred to an Office of Fair Trading report⁵¹ of 2011 into the UK equity underwriting market.
- Commenting that raising equity for single purpose businesses such as interconnectors is more expensive than for more diversified businesses.
- Referring to a 5% assumption in RIIO-ED2⁵² for the cost of raising new equity.

3.186 Four stakeholders expressed the view that there is insufficient evidence to support decreasing the 5% equity transaction cost allowance. Three stakeholders questioned the applicability of the CEPA report, with one cautioning that it is out of date due to drawing on data from 2011 (the Office of Fair Trading report on competition in the UK equity underwriting market). Two stakeholders contended that there has been insufficient evidence since the 2018 CEPA report to support lowering the 5% equity transaction cost allowance, with one asserting that introducing the change without sufficient evidence would damage the credibility of the cap and floor regime.

3.187 Two stakeholders also cautioned that lowering the equity transaction cost allowance would deter investment, with one specifically noting that it would deter investment and incentivise developers to minimise their costs during the early development phases of the project through not incurring de-risking costs, such as marine surveys.

3.188 A stakeholder commented that the potential equity transaction cost reduction (if the suggestion in the 2018 CEPA report referred to above were followed) “may tend to sustain the current concentration of TSO sponsored projects at the expense of independent developers”.

⁵¹ Office for Fair Trading/National Archives (2011), [OFT1303 Equity Underwriting and Associated Services | Office for Fair Trading/National Archives](#), of January 2011, page 21 onwards

⁵² Ofgem (2022), [RIIO-ED2 Final Determinations Finance Annex](#) UK asset classes, 1900-2023 from the 2024 DMS Database

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Decision and rationale

Financial parameter	Decision
Equity transaction cost	We have decided to retain the equity transaction allowance as 5% in the default regime.

3.189 The responses on this topic provided some evidence of the activities that are remunerated by the equity transaction cost allowance, albeit not comprehensively or in the quantitative detail that would provide more certainty of the relationship of this allowance and underlying costs.

3.190 We agree that more up to date evidence would be helpful in this area of equity transaction costs. This would be linked to more detail of the processes of equity raising for interconnector projects (and costs of specific activities in those processes).

3.191 We consider that the IDC allowance is the place in the cap and floor regime where development risk is reflected into the RAV and hence into the cap and the floor, rather than remunerating development risk being a purpose of the equity transaction cost allowance.

3.192 We may look to obtain further evidence on the activities remunerated by the equity transaction cost allowance in any future windows.

4. Summary of changes to financial parameters used to calculate the returns at the cap and the floor for Window 3 Interconnectors

- 4.1 In .
- 4.2 Table 7 below, we present an illustrative example of the returns at the cap for Window 3 Interconnectors. This illustrative example is based on a reference date of 30 September 2024. We compared these parameters to parameters calculated based on the same reference date and the methodology from Window 2.

Table 7: Return at the cap – comparison of Window 2 and Window 3

	Window 3	Window 2
RFR	1.39%	1.60%
TMR	6.75%	7.10% ⁵³
RPI adjustment	-	0.40%
Equity beta	1.125	1.25
Return at the cap (cost of equity)	7.42% (CPIH-real)	7.98% (RPI-real)

Source: Ofgem’s analysis

- 4.3 In the table below, we present an illustrative example of the returns at the floor for Window 3 Interconnectors. This illustrative example is based on a reference date of 30 September 2024. We compared these parameters to parameters calculated based on the same reference date and the methodology from Window 2.

Ofgem (2022), [RIIO-ED2 Final Determinations Finance Annex](#) UK asset classes, 1900–2023 from the 2024 DMS Database

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Table 8: Return at the floor – comparison of Window 2 and Window 3⁵⁴

Description	Window 3	Window 2
Return at the floor (cost of debt)	3.55% (CPIH-real)⁵⁵	2.08% (RPI-real)⁵⁶

Source: Ofgem's analysis of IHS Markit data

⁵⁴ Please note, the transition from RPI-real rates of return in W2 to CPIH-real rates of return in W3 should not have a material impact on nominal investor returns (over and above the impact of underlying debt cost movements). For further details on this, please see Inflation section on page 10.

⁵⁵ As shown in Table 1: Return at the floor calculations

⁵⁶ As shown in

Table 9: Rate of return at the floor (Window 2 methodology) below

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Table 9: Rate of return at the floor (Window 2 methodology)

Date	Index 1	Index 2	Average (nominal)	Inflation	Real Average Rate
20 days to FID	A 10yr+	BBB 10yr+	$\text{Avg} = (A + \text{BBB}) / 2$	10-year breakeven inflation	$= (1 + \text{Avg}) / (1 + 10\text{-year breakeven inflation}) - 1$
03/09/2024	5.29%	5.78%	5.53%	3.37%	2.10%
04/09/2024	5.25%	5.74%	5.50%	3.34%	2.09%
05/09/2024	5.25%	5.73%	5.49%	3.36%	2.07%
06/09/2024	5.23%	5.71%	5.47%	3.34%	2.06%
09/09/2024	5.25%	5.72%	5.49%	3.34%	2.07%
10/09/2024	5.21%	5.69%	5.45%	3.31%	2.07%
11/09/2024	5.16%	5.64%	5.40%	3.27%	2.07%
12/09/2024	5.19%	5.67%	5.43%	3.28%	2.08%
13/09/2024	5.18%	5.67%	5.42%	3.29%	2.06%
16/09/2024	5.17%	5.66%	5.41%	3.30%	2.05%
17/09/2024	5.18%	5.66%	5.42%	3.33%	2.02%
18/09/2024	5.24%	5.72%	5.48%	3.41%	2.01%
19/09/2024	5.28%	5.75%	5.52%	3.39%	2.06%
20/09/2024	5.30%	5.78%	5.54%	3.38%	2.09%
23/09/2024	5.32%	5.80%	5.56%	3.40%	2.09%
24/09/2024	5.33%	5.80%	5.57%	3.40%	2.09%
25/09/2024	5.37%	5.84%	5.61%	3.41%	2.12%
26/09/2024	5.40%	5.86%	5.63%	3.39%	2.16%
27/09/2024	5.39%	5.85%	5.62%	3.39%	2.15%
30/09/2024	5.39%	5.85%	5.62%	3.40%	2.14%

20-day average to FID (rounded)

2.08%

Source: IHS Markit, BOE